Question:

A 17-year-old teenager is brought to the emergency department with severe bleeding from his right hand. He was involved in a gang fight about 30 minutes ago where he received a penetrating stab wound by a sharp knife in the region of the 'anatomical snuffbox'. A vascular surgeon is called in for consultation. Damage to which artery is most likely responsible for his excessive bleeding?

Answer Choices:

A. Radial artery

B. Ulnar artery

C. Princeps pollicis artery

D. Brachial artery

Reasoning + Answer:

<think>

Finding Reasoning Paths:

1. **Anatomical Snuffbox** âĨĴ **Extensor Pollicis Longus Tendon** âĨĴ **Compartment** âĨĴ **Vascular Injury** âĨĴ **Radial Artery**.

2. **Anatomical Snuffbox** âĨĴ **Abscess** âĨĴ **Tendon Injury** âĨĴ **Compartment** âĨĴ **Vascular Injury** âĨĴ **Radial Artery**.

3. **Anatomical Snuffbox** âĨĴ **Tendon Injury** âĨĴ **Compartment** âĨĴ **Vascular Injury** âĨĴ **Radial Artery**.

Reasoning Process:

1. The anatomical snuffbox is an important anatomical landmark in the forearm. It is bordered by the **lateral aspect of the forearm**, **thumb**, and **lateral and medial digital expansions of the hand**. The **abductor pollicis longus** and **extensor pollicis longus** tendons pass through this region, and it is contained within the **second dorsal compartment** of the forearm.

2. A penetrating stab wound in the anatomical snuffbox can cause direct damage to structures in this compartment. The second dorsal compartment contains the **radial artery**, **radial nerve**, **extensor pollicis longus tendon**, and **abductor pollicis longus tendon**.

3. The radial artery is a major superficial artery of the forearm that runs through the dorsal compartment of the forearm and passes through the anatomical snuffbox. It is highly vulnerable to injury in this region due to its superficial location and anatomical course.

4. The **extensor pollicis longus tendon** and **abductor pollicis longus tendon** are also located in the second dorsal compartment and are less likely to be the primary source of bleeding, as they are muscles and tendons rather than blood vessels.

5. An **abscess** in this region would typically present with localized swelling, warmth, and infection rather than active bleeding. While an abscess could theoretically cause vascular injury indirectly, it is not the most direct or likely cause of excessive bleeding in this case.

6. The **radial nerve** is also located in the second dorsal compartment, but it is a nerve and does not bleed. Therefore, it is unlikely to be the source of the bleeding.

7. Based on the anatomical location of the injury (anatomical snuffbox) and the structures contained in the second dorsal compartment, the most likely source of bleeding is the **radial artery**.

Conclusion:

The damage to the **radial artery** is the most likely cause of the excessive bleeding.

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<answer>

Radial artery

</answer>